

Troubleshooting

Every entry below is symptom -> check -> fix, grounded in the code that ships in this repo, not in how a hook is supposed to behave. Where a command is shown as `<crew>/hooks/scripts/...`, that is `${CLAUDE_PLUGIN_ROOT}/hooks/scripts/...` inside a command file, or the plugin's cache directory when you run it by hand — see the shorthand in [Memory and Obsidian](#).

Stale Install

Symptom: a command from this guide (`/crew:status`), a new hook, a config key) does not exist, or behaves like an older release.

- **Check the installed version against the repo.**

```
cat ~/.claude/plugins/installed_plugins.json | grep -A3 '"crew"'
```

Compare the commit or version it records against `plugin/crew/.claude-plugin/plugin.json`'s version in the marketplace clone. `claude plugin update` decides whether to re-copy a plugin by comparing the **declared version**, not its content — a content change that ships with no version bump means every already-installed copy reports "already at the latest version" forever, holding the old code, with nothing in the repo itself looking wrong (`CLAUDE.md`, "Stop and ask"). This class of bug is why `/crew:status`'s absence, or a hook that clearly predates a documented fix, is worth checking here before assuming the fix is wrong.

A related, sharper case: on a `directory-source` install (a dev checkout registered as a local plugin path rather than pulled from a marketplace), `installed_plugins.json` can name a version string with **no relationship to the code that actually executes** — the tree runs whatever is on disk regardless of what that file says (`TOD0.md`, "`installed_plugins.json` records 0.19.56 while 0.19.63 executes"). On that install shape, trust the code over the version string.

- **Fix:**

```
claude plugin update crew@useful-claude-add-ons
```

then **restart** `claude` — a running session does not pick up a re-copied plugin. If the update reports "already at the latest version" and you can see the fix in the marketplace repo's `main`, the version was not bumped when it should have been; that is a defect to file against the plugin, not

something a client-side retry fixes.

- **Check it worked:**

```
/crew:status
```

Read-only, at most 40 lines, dispatches nothing. Its `config` line names which config file it read (`.crew/crew.json` for 1.0, `.crew/config.json` for 0.20); codemap reports behind: `<subsystem>` when a map's anchor and the path diff both say it is stale.

Hook Noise

Every hook crew registers, from `plugin/crew/hooks/hooks.json`, and the config key that turns its behaviour up or down. All events are registered once for bash and once for PowerShell (`shell: "powershell"`), so a Windows machine with both bash and pwsh on PATH fires both halves of every pair — that is what "even counts" in the file means, not a bug.

Event	Script	What it does	Switch
SessionStart	handoff-read.sh	resets its once-per-session markers; reads back <code>.work/HANDOFF.md</code> only when <code>memory.inject</code> is false	— (always runs)
SessionStart	platform-sync.sh	detects/repairs <code>platform.{os,wsl,shell,windows,HostIp}</code>	— (always runs)
SessionStart / UserPromptSubmitted / PostToolUse / SubagentStart	crew-context.sh	injects code-map slices and vault recall, tracks the token budget	<code>context.enabled</code> (whole hook), <code>memory.inject</code> (codemap, handoff and vault injection — on by default since 1.0.0)

Event	Script	What it does	Switch
UserPromptSubmit	approval-hook.sh	records or refuses a /crew:approve <id> receipt	always records; enforcement depends on scope.mode
PreToolUse (Bash/PowerShell)	promote-gate.sh	the six command/production guards plus mergeGate	guards.terraformApply, guards.forcePush, guards.adminMerge, guards.mergeGate, guards.cloudDestructive, guards.sqlDestructive, guards.prodDatabase, guards.prodServer
PreToolUse (Write Edit)	role-write-guard.sh	refuses a write outside the dispatched role's declared scope	guards.roleWrites (block/report/off)
PreToolUse (Bash PowerShell)	cloud-guard.sh	destructive aws/az commands, wrong- identity commands, and dispatches of workflows listed in environments.workflow s (guards.deployWorkflow) — judged only when every word on the line is a plain literal or a single-quoted word; anything else asks, and is refused unattended	guards.cloudGuard (block/report/off)
PreToolUse (Bash PowerShell)	cloud-guard.sh, environment layer (T-0005)	a terraform apply judged by its target environment, and a destroy never applied	environments.nonProd (repo only; globs naming non- production workspaces), environments.prodUnattended

Event	Script	What it does	Switch
		unattended	(both layers; true only when both say true)
PreToolUse (Write Edit MultiEdit NotebookEdit Bash PowerShell)	scope-guard.sh	plan-approval + ticket scope guard	scope.mode (off/report/block/auto), scope.allowCliApproval
PreCompact	handoff-write.sh	writes the handoff note before compaction	context.autoWrapUp, context.handoffPath
Notification	notify.sh	pings an external channel	notify.provider (none disables it), notify.events
Stop	verify-gate.sh	runs .crew/verify.json's checks	verifyGate (boolean)
Stop	context-watch.sh	nags for a handoff near the context budget, drives auto-clear	context.enabled, context.warnAt, context.budgetTokens, context.reserveTokens, context.autoClear.*
Stop	completion-audit.sh	diffs the whole tree against the ticket's scope base	scope.mode

- **Silence one hook without touching the rest:** set its own key. `guards.roleWrites: off`, `guards.cloudGuard: off` and `scope.mode: off` are already the shipped defaults — a noisy session usually means one of these was turned on somewhere (repo or machine-global) and forgotten, not that the default changed.
- **Silence the SessionStart context specifically:** `{"memory": {"inject": false}}` stops the context hook; `handoff-read.sh` then prints the handoff on resume instead.
- **See what the context hook is actually doing:**

```
python3 "<crew plugin root>/hooks/scripts/crew_context.py" --stats
```

Reports emissions, injected characters, dedup hits, and vault-recall hit/miss/skip counts with a reason breakdown (see "Obsidian" below). The log itself is `<git-common-dir>/crew/context-log.jsonl` (or `.work/crew/context-log.jsonl` outside git), shared across every worktree of the repo. - **A hook fails and you cannot tell why:** every wrapper here resolves its own Python (python3/python/py) rather than assuming one is present, and every one that can block fails **closed** — never silently — when no interpreter resolves. See "no-python fail-closed" under Scope, below; the same shape applies to `scope-guard.sh`, `role-write-guard.sh`, `cloud-guard.sh` and `verify-gate.sh`.

Review Loops

Each ticket gets **two** review rounds, reserved before the reviewer runs, tracked in `<git-common-dir>/crew/review/<ticket>.json` — shared by every worktree of the repo, so a second worktree of the same repo spends the same budget (`review_ledger.py`).

- **Symptom: review keeps re-running / "budget exhausted". Check:**

```
python3 "<crew>/hooks/scripts/review_ledger.py" --root . --ticket <id> --status
```

A round's outcome is CLEAN, FINDINGS or INCOMPLETE. A completed round 2 — either FINDINGS or INCOMPLETE — leaves the ticket state REVIEWED, so its FINDINGS can still be accepted. A **third** reservation attempt is refused outright and the state becomes NEEDS_REPLAN; that refusal, and an explicit `--reject`, are the only two ways into NEEDS_REPLAN. **Fix:** own the FINDINGS with `--accept --by <who>` (only the most recent completed round, only once, never once NEEDS_REPLAN), or write a new plan and get it approved — `crew_ticket.py approve` on a NEEDS_REPLAN ticket opens a fresh budget of two rounds counted from the successor plan; the rounds already spent stay in the ledger and are not erased.

- **Symptom: /crew:done refuses with a receipt error. Check:**

```
python3 "<crew>/hooks/scripts/review_ledger.py" --root . --ticket <id> --check-receipt
```

A CLEAN verdict writes a receipt automatically; a FINDINGS verdict only becomes one through `--accept`. `--check-receipt` rebuilds the review bundle from the receipt's recorded base and fails

unless the hash still matches, the receipt is for the **latest** recorded round, and the state is not `NEEDS_REPLAN` — so editing a file after the reviewer read it, or after the receipt was written, invalidates the receipt even though nothing about the ledger itself looks wrong. **Fix:** if the edit was deliberate, get the ticket reviewed again (spends the next round); if it was accidental, revert the edit and re-check.

- **Symptom: you want to send a ticket back without spending the third round.**

```
python3 "<crew>/hooks/scripts/review_ledger.py" --root . --ticket <id> --reject --by <who>
```

Refuses on a ticket already `ACCEPTED` or already `NEEDS_REPLAN`.

Scope: Approval and the Completion Audit

A ticket's contract lives in `.work/tickets/<id>/: spec.md's ## Touch` section names every path the ticket may change; `plan.md's Files:` lines must each fall inside `Touch`. Two hooks hold a session to that contract — see [Daily workflow: scope and approval](#) for the contract itself. This section is what goes wrong with the approval and the audit.

- **Symptom: an edit inside Touch is still refused. Check:** approval status.

```
python3 "<crew>/hooks/scripts/crew_ticket.py" status --ticket <id>
```

Returns `approved` (both `spec.md` and `plan.md` hash to what was approved), `stale` (either file changed since), or `none`. **Editing `spec.md` or `plan.md` after approval makes the approval stale immediately** — that is deliberate: an edited spec cannot silently widen what the guard accepts. Only a receipt from the user's own prompt counts by default; a `cli`-sourced receipt (written by `crew_ticket.py` `approve` directly, for tests/CI) is accepted only when `.crew/config.json` sets `scope.allowCliApproval: true`. **Fix:** re-approve. The user types `/crew:approve <id>` again — nothing else can write the receipt; `scope_guard.py` refuses a `Write/Edit` under `<git-common-dir>/crew/` in every mode but `off`, so a session cannot forge or refresh its own approval.

- **Symptom: you need to touch one more path mid-ticket. Fix:** amend `spec.md's ## Touch` (widen it), then approve again. There is no partial-approve; amending scope is edit-then-approve, same as any other spec change.

- **Symptom:** `/crew:done` refuses on "out of scope" for a file the edit guard never saw.

This is the Stop-time completion audit (`completion_audit.py`), not the edit guard. The edit guard only sees `Write/Edit/MultiEdit/NotebookEdit` tool calls; a `sed -i`, a redirect, a formatter or a `git mv` never reaches it. The audit instead diffs the **whole working tree** against the ticket's scope base (`scope_base.resolve` — the commit the ticket started from) across committed, staged, unstaged and untracked changes, so a shell-made write is caught here even though nothing blocked it at the time. **Check it directly, without waiting for a Stop:**

```
python3 "<crew>/hooks/scripts/completion_audit.py" --check --ticket <id>
```

Fix: either the path genuinely needs to be in scope (widen Touch and re-approve), or revert the out-of-scope change. Files `git` ignores (including everything under `.crew/`) and `.work/` itself are outside what the audit can see at all — that is by design, not a gap to work around.

- **scope.mode values, and what "auto" means:** `off` (hooks do nothing, the default), `report` (allows everything, logs the row to `.crew/guard.log`), `block` (refuses out-of-scope writes and fails the completion audit), `auto` (report for the first ten tickets approved in this repo, then `block`). A config file that exists but fails to parse, or names an unknown mode, resolves to `block` and says so — the unknown never collapses into the permissive value.
- **No-python fail-closed, and why it is not a bug:** if none of `python3/python/py` resolves to a *proved* working interpreter, `scope-guard.sh` (and `completion-audit.sh`, byte-for-byte the same logic) does **not** silently allow the write. It checks whether `.crew/config.json` **provably** sets `scope.mode` to `off` — a crude, non-JSON textual check, because it runs in exactly the situation where `python` cannot parse the file for it — and only then exits 0. Anything it cannot prove `off` (a corrupt file, `report`, `auto`, `block`, or no `scope` key at all) fails **closed** with exit 2: "no usable python ... failing closed". Fix by installing a real Python 3, not by reading the closed refusal as a false positive.

Cloud Guard False Positives

`cloud-guard.sh` / `cloud_guard.py`, a `PreToolUse` hook on Bash and PowerShell. **Ships off** (`guards.cloudGuard: off`); turn it on with `{"guards": {"cloudGuard": "report"}}` first to see what it *would* refuse before enforcing with "block".

- **SQL text that merely looks destructive is not blocked.** `DROP/TRUNCATE` inside a string

literal or a comment does not count: the guard strips literals and comments before matching, read both as standard SQL and as MySQL, because the two dialects disagree about what `\'` and `--` mean — a payload is flagged only if **either** reading still shows the keyword outside a literal. So `psql -c "SELECT 'DROP TABLE x' AS s"` and `psql -c 'SELECT 1 -- DROP TABLE x'` are both allowed (plugin/crew/tests/test_cloud_guard.py, cases sql-literal/sql-comment/sql-block-comment). If a query like this is still blocked, the reason (`[sqlDestructive]` in the deny message) is worth reading closely — the keyword may genuinely sit outside every literal reading.

- **A dry-run, help, or -WhatIf invocation is still classified as destructive.** The guard matches on the *subcommand*, not on flags that make it a no-op: `terraform destroy --help`, `aws ec2 terminate-instances --dry-run` and `Remove-AzResourceGroup -WhatIf` are all judged exactly like the command they simulate, because nothing in `_terraform_destructive/_aws_destructive/the Remove-Az*` matcher inspects those flags. This is a known gap, not a config bug — there is no key that special-cases them. If you need to run one of these unattended, use report mode while testing, or approve the one-shot marker the deny message names. A literal `gh workflow run ... --help` (or `-h`) is the exception: `gh` prints help and dispatches nothing, so the line is not judged at all; `terraform's --help` still is.
- **A workflow dispatch is refused as "could not tell" although it looks fine.** With `environments.workflows` set, a `gh workflow run` or `gh api .../dispatches` line is judged only when every word on it is a plain literal (letters, digits, `_`, `./`, `:=@%+`, `-`) or one whole single-quoted word, joined only by `;`, `&&`, `||`, `&`, newlines, `>/>>/&>/&>>` to a plain word and `2>&1`. A pipe, any `<`, double quotes, `$`, a glob, `bash -c '...'`, `xargs`, a command word built at run time (`$X $Y run ...`, `Start-Process $x`, an alias to a variable), `gh` reading stdin or a file (`--json`, `--input`, `-F k=@f`), and a malformed `environments` block in either config layer make the line could-not-tell: it asks when you are there and is denied unattended at every setting. **Fix:** write the literal form — `-f environment=staging` instead of a `--json` body, `'Deploy Staging'` instead of `"Deploy Staging"`, `> log` instead of `| tee log`, the value instead of `$ENV` — or approve the one command with the marker the refusal names (it covers those exact bytes only). Other `gh` commands (`gh pr create --title "..."`) are never gated.
- **An "unpinned" repo makes an ordinary destructive command ask or deny, even when it is obviously fine.** With no `cloud.*` pins, a read-only command (`aws s3 ls`, `az group list`) runs unchecked — but *any* destructive command has an identity the guard

cannot vouch for, which is **unknown**, and unknown is never allowed unattended: it **asks** when a person is there to answer and is **denied** when not (CREW_UNATTENDED=1, CI, or a non-interactive session). This is the rule most likely to surprise a first-time user, because it fires even in a repo that has never configured anything cloud-related. **Fix — pin the identity this repo may act as**, in `.crew/config.json` (repo-only; a machine-global cloud block is ignored):

```
{ "cloud": {  
  
  "awsProfiles": ["acme-dev", "acme-sandbox-*"],  
  
  "awsRegions": ["eu-west-1"],  
  
  "azureSubscriptions": ["00000000-0000-0000-0000-000000000000", "Acme Dev"] } }
```

A profile/region/subscription outside the pins is denied outright ([cloudIdentity]). Static AWS_ACCESS_KEY_ID credentials and Remove-Az*'s saved PowerShell context are **always** unknown — the guard has no way to read either — so pinning does not silence those; only removing the static keys or naming the account through `--profile/--subscription` does.

Obsidian

See [Memory and Obsidian](#) for setup. What goes wrong day to day:

- **Symptom: nothing is being captured / gardened. Check:** is the *primary* vault reachable? Capture, import and the gardener write to the primary vault only, and **never** fall back to a recall vault. If the primary is unavailable — an unmounted drive, a renamed folder — they write nothing and say why on stderr; the session itself is never blocked. `$v0 adopt` shows which vault is currently primary.
- **Symptom: recall never surfaces anything you know is in the vault. Check:**

```
python3 "<crew plugin root>/hooks/scripts/crew_context.py" --stats
```

Read the `miss reasons` line. `cli-missing` means `obsidian-vault` is not installed where `crew` can find it; `cli-exit-N`, `cli-timeout`, `cli-bad-json` mean the CLI ran and failed; `no-vaults` means no vault is currently primary or recall; `no-hits` means the search itself found nothing. Only `no-hits` means "the vault genuinely has nothing relevant" — every other reason is a setup problem. Also

confirm `memory.inject` is not false in `.crew/config.json` — recall and code-map injection are on by default since 1.0.0, so "no recall ever appears" with the key set to false is expected. **Fix:** `$V0 recall --query "<words you expect>"` directly, to separate "the CLI can't find it" from "the hook isn't calling the CLI".

- **Symptom: the gardener seems to be falling behind.** It runs in bounded passes on purpose, not continuously: **at most 5 items or 10 minutes per run**, whichever comes first, on **one designated host**, one run at a time. A session whose transcript lives on another machine is left queued for that machine and does not spend one of the five slots. **Check:** `$V0 queue` (what is waiting) and `~/.claude/obsidian/gardener.log` for lines starting `left queued`. **Fix:** run a bounded pass by hand (`/obsidian-vault:garden` or `$V0 garden-run`), or drain a large backlog in batches: `$V0 drain --apply --batches 4`.
- **Symptom: two machines sharing one vault seem to be racing each other.** They are not writing the same file: each host appends to its own `inbox/pending-reflect.<host>.md` and acknowledges to its own `inbox/reflected.<host>.md`, so two machines syncing one vault never contend for the same queue file. An older `inbox/pending-reflect.md` (no host suffix) is still read if present, so an old backlog drains rather than being stranded — it is just never written again.
- **Symptom: `/crew:brainstorm` or `/crew:fix` stops because create refused a ticket id.** On the Obsidian ticket board, `crew_tracker.py create` refuses an id it cannot prove is yours, and prints one of four lines (paths and ids here are examples). The first three begin id taken, and brainstorm and fix then take the next free id on their own. For any other refusal they stop and show you the line.
 - **Another repo owns it.** `obsidian: could not update: id taken: T-0060` on `Board.md` belongs to another repo (`https://example.invalid/team/app`, per `T-0060.md`), not this one (`https://example.invalid/other/app`): give this repo its own `obsidian.boardDir` **Fix:** two repos share one board. Give each repo its own `obsidian.boardDir`.
 - **Nobody can tell who owns it.** `obsidian: could not update: id taken: could not tell whose card T-0042 is` (no `T-0042.md` note names its repo-id); if it is this repo's, put `'repo-id: https://example.invalid/other/app'` in `T-0042.md` **Fix:** if the card is yours, add the line it names to the note (see "The ticket board" in [Memory and Obsidian](#)). If it is not yours, let brainstorm take the next id.
 - **Another session in this repo holds it.** `files: could not update: id taken: T-0050 is already ready "Fix tests"` **Fix:** none needed; `.work/INDEX.md` already

has that ticket. This check runs before the vault is opened, so it wins even when the vault is missing.

- **Crew cannot tell which repo this is.** obsidian: could not update: could not tell this repo's identity: git gave neither an origin URL nor a common dir for <repo path> **Fix:** run crew inside a git checkout. `git rev-parse --git-common-dir` must answer there.

Auto Wrap-Up, Clear and Resume

See [Auto wrap-up, clear and resume](#) for the full cycle that lets a long session finish itself cleanly, clear, and pick up where it stopped — what wrap-up, auto-clear and resume each do, the machine-global keys that turn auto-clear on, and every reason a clear can refuse (logged to `.crew/.autoclear.log`).

"Auto-Clear Did Nothing on Windows"

Symptom: auto-clear is on, a handoff is written and verified, and nothing types the configured command — no keystrokes appear at all.

- **Check `context.autoClear.method`** with `/crew:config --explain` or `crew_config.py --explain`. On native Windows with no tmux pane, "auto" (the default) resolves to "notify", not to any keystroke method — **this is not a bug**, it is the 1.0 owner decision: auto never arms SendKeys on your behalf. "notify" types nothing at all; instead the Stop hook prints a `systemMessage` telling you the handoff is written and verified and it is safe to run the configured command yourself.
- **If you want keystrokes typed for you**, ask for `sendkeys` by name — it is opt-in only, and `/crew:init`, `/crew:migrate` and `/crew:onboard` all refuse to write it without an explicit yes. It declines and falls back to `notify` (logged to `.crew/.autoclear.log`) inside Windows Terminal specifically, because that host puts every tab in one window and nothing outside it can confirm which tab is active — this is also not a bug, and switching terminal hosts is the only fix.
- **Fix:** either treat the `notify` message as the intended behaviour, or run `crew_autoclear_setup.py apply-method sendkeys --yes` (`${CLAUDE_PLUGIN_ROOT}/hooks/scripts/`) to opt in, after reading what SendKeys does. See `plugin/crew/CONFIG.md §14`.

"Claude Code Compacted by Itself"

Symptom: the session cleared or summarised itself with no `/clear` or `/compact` typed, and no message about auto-clear ran first.

That is Claude Code's own **built-in auto-compact**, not this plugin. Crew neither causes nor tunes it — `context.autoClear` and `context-watch.sh` only ever act *after* a handoff is written and verified, and nothing in this plugin can trigger a compaction on its own. There is no crew setting that changes when Claude Code's auto-compact fires.

The one thing crew does around either kind of reset: on the next `SessionStart` after any `/clear` or `/compact` — self-initiated or Claude Code's own auto-compact — `handoff-read.sh` reloads `.work/HANDOFF.md` back into context automatically, so a compaction you did not ask for still resumes from the last written handoff rather than from nothing.

Turning Things Off

Every switch named above, in one place. "Off" for a guard means the `PreToolUse` hook still fires but returns immediately without judging anything; "off" for `verifyGate` means the `Stop` hook does nothing at all that turn.

Key	Lives in	Values	What "off"/floor means
<code>memory.inject</code>	repo only	bool	false: no code-map, handoff or vault text is injected (default true since 1.0.0)
<code>guards.cloudGuard</code>	both layers, ratchets	block/report/off	off: the hook reads this key and exits
<code>guards.roleWrites</code>	both layers, ratchets	block/report/off	off: every Write/Edit is allowed unconditionally
<code>guards.terraformApply,</code>	both	block/	there is no "off" — allow is the most

Key	Lives in	Values	What "off"/floor means
forcePush, adminMerge, mergeGate, cloudDestructive, sqlDestructive	layers, ratchets	ask/allow	permissive tier, still logged
guards.deployWorkflow	both layers, ratchets	block/ask/allow	block is default and floor; allow covers nonProd only — production without environments.prodUnattended in both layers, and an unknown environment, still ask (denied unattended)
environments.nonProd	repo only	list of globs	[]): nothing is nonProd, so every terraform apply asks (denied unattended)
environments.prodUnattended	both layers, ratchets (true only when both say true)	bool	false in either layer holds it down: production applies and dispatches ask
guards.prodDatabase, guards.prodServer	both layers, ratchets	none/read/full	none is both the default and the floor
scope.mode	repo only	off/report/block/	off: neither the edit guard nor the completion audit runs

Key	Lives in	Values	What "off"/floor means
		auto	
<code>scope.allowCliApproval</code>	repo only	bool	false: only a /crew:approve typed by the user counts
<code>verifyGate</code>	repo only	bool	false: the Stop verify gate does not run at all
<code>context.enabled</code>	both layers	bool	false: context-watch.sh (the Stop nag) does nothing
<code>context.autoClear.enabled,</code> <code>autoWrapUp</code>	both layers	bool	each independently disables one leg of the wrap-up/clear loop
<code>notify.provider</code>	both layers	provider name / none	none: the Notification hook sends nothing
<code>change.requireForProduction</code>	both layers, ratchet s (may only turn ON)	bool	false: promoting needs no approved change request
<code>install.policy</code>	both layers, ratchet s	see plugin/crew/CONFIG.md §"install"	narrowest tier refuses more install actions

A destroy is never applied unattended, at any setting: `destroy`, `apply -destroy`, `-replace`, `workspace delete`, a saved plan whose sidecar lists a `delete`, or an `apply` with no sidecar at all asks when you are there and is denied unattended — `terraformApply: allow` and `environments.prodUnattended` included. The full decision table is in `plugin/crew/CONFIG.md` (`environments.*`).

~/.claude/crew/config.json is the machine-global layer; .crew/config.json is per-repo and wins where both speak. A **ratcheted** key (marked above) can only be *narrowed* by the repo relative to the machine-global value, never widened — a repo cloned from someone else cannot silently loosen a guard the machine owner set to block. /crew:config shows where each setting actually came from.

Web Testing

Symptom: `--select web-testing` (or the ticked default row) fails, or Playwright runs but the visual rule never passes.

- **Node too old.** `install-prerequisites.{sh,ps1}` checks for Node ≥ 20.19 (or $= 22.12$) itself and refuses rather than installing or upgrading Node for you — `node -v`, install a newer one, and re-run `--select web-testing`.
- **Browsers missing system dependencies.** `npx playwright install --with-deps chromium` is what the installer runs; a bare `npx playwright install` (no `--with-deps`) skips the OS packages Chromium needs and produces a browser that launches, then crashes on first navigation.
- **Docker absent -> visual UNVERIFIED, not a fail.** `toHaveScreenshot` baselines are only meaningful when generated inside `mcr.microsoft.com/playwright:v1.63.0-noble` — the installer warns rather than blocking when Docker is missing, and the `visual.verify.json` rule is written to skip (report UNVERIFIED) outside that image rather than compare against a baseline that can never match.
- **MCP server won't connect on Windows.** Both `@playwright/mcp` and `chrome-devtools-mcp` are launched with `npx`, which the MCP host cannot invoke directly on Windows — wrap the entry as `"command": "cmd", "args": ["/c", "npx", ...]` in `.mcp.json` (documented in `chrome-devtools-mcp`'s own troubleshooting for the "Connection closed" failure); the `stack-web` skill's `.mcp.json` example already shows this.

Windows Notes

- **pwsh is not on Git Bash's PATH.** Every script that shells out to PowerShell names it by an absolute path or resolves it explicitly; a bare pwsh in a hand-typed command fails as "command not found", which is easy to misread as PowerShell itself being broken.
- **Git Bash ships without python3.** Every hook wrapper here resolves python3, then python, then py, and fails loudly on stderr rather than silently standing down when none resolves (see "no-python fail-closed", above). If a hook you expect to run says nothing at all, check for a usable Python on PATH first.
- **Both flavours are always registered.** hooks.json pairs every bash command with a shell: "powershell" sibling on the same event, so a machine with both bash and pwsh present runs both. This is not double-firing to fix — it is how the same hook reaches both shells; each PowerShell twin stands itself down on the wrong platform (`$env:OS -ne 'Windows_NT'`) rather than the bash twin doing that job.

Auto Wrap-Up, Auto-Clear and Auto-Resume

This section is folded into the troubleshooting guide (troubleshooting.md), linked from it as "Auto wrap-up, clear and resume". It covers the cycle that lets a long session finish itself cleanly, clear, and pick up where it stopped.

What the Cycle Does

1. **Wrap-up.** When the context window crosses its threshold, the Stop hook blocks once. It asks the session to finish or park the change in flight, write `.work/HANDOFF.md`, and update the ticket. This happens once for each threshold crossing in each session. It never happens on the forced continuation that follows a block.
2. **Clear, or notify.** After the handoff is written, auto-clear acts on it. What it does depends on method (below): `notify` (the default on native Windows) prints a message telling you the handoff is written and it is safe to run `/clear` (or `/compact`) yourself — it types nothing. `tmux`, `xdotool` and `sendkeys` actually type the configured command into the terminal that runs this session. Whichever method is in force, it acts only when all four of these hold: - this machine opted in; - the handoff was written by this session after the wrap-up request, is longer than a stub, and is not the automatic PreCompact skeleton; - the context reading that caused the wrap-up was a real measurement (see "When it will

not clear"); - for a method that types (tmux/xdotool/sendkeys), the target window can be identified exactly — notify needs none of this, since it identifies no window.

Crew neither causes nor tunes Claude Code's own auto-compact. That is a separate, built-in behaviour that fires whenever Claude Code itself decides to; this cycle only reacts to a handoff once one exists, and nothing here makes auto-compact happen sooner, later, or at all. 3.

Resume. The new session starts with the handoff and its next action already in context, within the 3,000-character resume budget. Nothing starts working unprompted: crew does not use `initialUserMessage`, so you press Enter to continue. Claude Code 2.1.282 drops it in an interactive session anyway (spike, 2026-09-25): it is honoured by `claude -p` only.

With `resume.auto: true` in the machine-global config, the handoff after `/clear` or a manual `/compact` also names the exact next command, read from the note's `resume: line` (`resume: /crew:done T-0001`): Auto-resume: ready to run `/crew:done T-0001`. — or Auto-resume did not start: `<reason>`. when a check refuses it. It still does not start on its own: you press Enter or type it, until T-0013 types it where the terminal can be identified. The refusal reasons are: compact was not a manual `/compact`; no handoff note, or it was archived as stale, or is stale and could not be archived; the handoff is the automatic PreCompact skeleton; no resume line, `resume: none`, or a line the grammar refuses (a second line, trailing text, an unknown command, or an excluded one such as `/crew:approve`); the note's `branch:` or `head:` does not match the checkout; the ticket directory or goal file is missing; the command is not installed; `handoff-author.json` could not be read; no record of which session wrote this handoff; the handoff changed since its author session wrote it; the handoff was written by another session; this session's process could not be identified (always on a host without `/proc`, such as native Windows or macOS); the record of past auto-resumes (`resume-state.json`) could not be read, or its directory cannot be searched; this handoff was already resumed; the progress fingerprint could not be computed; the same command with no progress since the last auto-resume; `internal error` (the decision itself failed, and the handoff is still injected). Never on a plain startup.

A note resumes only in the session that wrote it. Write it with the Write tool (as `/crew:handoff` does): the context hook then records which session and which Claude Code process wrote it. After `/compact` the session must match; after `/clear` the process must, so a `/clear` in another terminal on the same worktree waits. A note written by Bash or by hand waits.

Turning It on and Off

Wrap-up and resume are on by default. Auto-clear is **off** by default. You turn it on per machine,

in the machine-global config, because it drives this machine's keyboard:

Where	Key	Value	Effect
~/ .claude/ crew/ config.json (Windows: %USERPROFILE% \ .claude\crew\ config.json)	context.autoClear.enabled	true	turns auto-clear on for every crew repo on this machine
same	context.autoClear.method	"auto" (default), "tmux", "xdotool", "notify", "sendkeys", "none"	how the cycle acts — see "Which method, and what it does" below
same	context.autoClear.windowTitle	a substring of the terminal's title	fallback when the window cannot be found by process id
same	context.autoClear.delaySeconds	3	wait before typing, so the prompt exists
same	context.autoClear.minHandoffLines	5	a shorter handoff counts as a stub
.crew/ config.json (the repo)	context.autoClear.enabled	false	switches auto-clear off in this repo. A repo can switch it off, but true here does not switch it on.
~/ .claude/ crew/ config.json	resume.auto	true	names the next command from the handoff's resume: line

Where	Key	Value	Effect
			after <code>/clear</code> or a manual <code>/compact</code> (off by default)
<code>.crew/crew.json</code> or <code>.crew/config.json</code> (the repo)	<code>resume.auto</code>	<code>false</code>	vetoed auto-resume in this repo. <code>true</code> here does not switch it on.
<code>.crew/config.json</code>	<code>context.autoWrapUp</code>	<code>false</code>	replaces the wrap-up instruction with a plain handoff request. It still blocks once.
<code>.crew/config.json</code>	<code>context.enabled</code>	<code>false</code>	turns off the whole context watcher, wrap-up included
<code>.crew/config.json</code>	<code>memory.inject</code>	<code>false</code>	handoff - read prints the handoff for you to read, without extracting the next action.
			<code>context.autoResume</code> is no longer read.

`/crew:config` shows where each value comes from and walks you through the global file. To turn the cycle off on a machine, set `context.autoClear.enabled` to `false` or delete the key.

Arming one scratch repo while other sessions are live

`enabled: true` arms auto-clear in **every** Claude session on the machine, not just the one you are testing. If other sessions are running, one of them can get `/clear` typed into its terminal when its own handoff lands. To limit auto-clear to one repo or one session, add these keys to the same machine file. They can only narrow what `enabled` turns on:

Key	Value	Effect
<code>context.autoClear.onlyRepos</code>	list of absolute repo paths	auto-clear runs only in these repos
<code>context.autoClear.onlySessions</code>	list of session ids	auto-clear runs only in these sessions

- A missing key or `null` does not narrow anything. An empty list `[]` turns auto-clear off everywhere. If a key holds anything other than a list, auto-clear is off everywhere too.
- When both keys are set, a session must match both.
- Paths are compared after symlinks are resolved. Backslashes and trailing separators do not matter. On Windows, case does not matter and `/c/repos/x` means `C:\repos\x`. A relative path such as `.` never matches.
- Session ids must match exactly, including case.
- These keys are read **only** from the machine file. If a repo's `.crew/config.json` sets them, they are ignored, so a repo cannot add itself. A repo's `enabled: false` still turns auto-clear off.
- In any other repo or session, auto-clear stays silent. It does not write a log line, just as it does when `enabled` is off.

To arm one scratch repo safely:

```
{
  "context": {
    "autoClear": {
      "enabled": true,
      "onlyRepos": ["C:\\repos\\autoclear-scratch"]
    }
  }
}
```

On Linux, use a path such as `"/home/you/autoclear-scratch"`. When you are done testing,

remove `enabled` or set it to `false`. Also delete `onlyRepos`, so that a later `enabled: true` does not arm only that repo without anyone noticing.

Which Method, and What It Does

Method	What it does	Where auto picks it
<code>tmux</code>	Types the command into the <code>\$TMUX_PANE</code> that is an ancestor of the hook. No focus involved — the most reliable method.	Linux, macOS, WSL, Git Bash on Windows, when <code>\$TMUX</code> names a pane and <code>tmux</code> is on <code>PATH</code>
<code>xdotool</code>	Types into the one X11 window owned by the nearest ancestor process (or matching <code>windowTitle</code>).	Linux with <code>xdotool</code> on <code>PATH</code> and <code>\$DISPLAY</code> set
<code>notify</code>	Types nothing. Prints a message saying the handoff is written and verified and it is safe to run the command yourself. Never says anything was cleared or compacted — it does not know that, and it did not do it.	native Windows , whenever there is no <code>tmux</code> pane — this is the default there, not a fallback for something missing
<code>sendkeys</code>	<code>System.Windows.Forms.SendKeys</code> against the one window owned by an ancestor process (or matching <code>windowTitle</code>), confirmed still in the foreground right before typing. auto never picks this — typing into a window this hook found itself is a risk <code>auto</code> does not get to accept on your behalf. Set <code>method: "sendkeys"</code> explicitly to opt in. Even then, it declines and falls back to <code>notify</code> — logging why — when the target window is owned by Windows Terminal, because that host puts every tab in one	never (opt-in only)

Method**What it does****Where auto
picks it**

window and nothing outside it can confirm which tab is active.

none

Refused outright.

never

When It Will Not Clear (or Notify)

Each refusal is logged with its reason to `.crew/.autoclear.log`. A Stop hook's stderr is not shown to you, so check that log first. A notify or sendkeys decline is logged the same way — "declined" rather than "refusing", since the machine has opted in and something still happened (a message, or a fallback), just not a keystroke.

Log says**Meaning**

not trustworthy (estimated-from-transcript-size)

no usage record was found, so the reading was an estimate

not trustworthy (unknown-window)

the model's context window is unknown and `context.budgetTokens` is not set

not trustworthy (stale-reading-before-compaction)

the last reading came from before a compaction

the wrap-up marker is empty or not JSON

the marker was not written by this version of the hook. Nothing clears on it.

predates this session's wrap-up request

the handoff on disk is older than the request, probably left by an earlier session

automatic PreCompact skeleton

only the compaction skeleton exists, not a real handoff

N windows have a title containing ... / has N windows

the window is ambiguous, so nothing is typed (sendkeys/xdotool only)

no window belongs to any ancestor of this hook

the terminal's window could not be found by process id, and no `windowTitle` is set

Log says

Meaning

tmux pane %N could not be confirmed

(sendkeys/xdotool only)

\$TMUX_PANE is not the pane that runs this session

method wtype ... is refused

Wayland's wtype cannot target a window. Use tmux instead.

sent - method notify

the message was printed — this is the normal, successful outcome for notify

declined sendkeys - cannot verify the active tab ...

the target window is owned by Windows Terminal; sendkeys fell back to notify instead of guessing which tab is active

declined sendkeys - the target window lost focus during the ...s delay

the window had focus when identified, but not when the delay ended, so nothing was typed

Two sessions in one repository no longer interfere with each other. Wrap-up markers are `.crew/.handoff-requested-<session_id>`, and a `SessionStart` clears only its own session's marker.

Which Window Gets the Keystroke

Applies only to the methods that type — `tmux`, `xdotool`, `sendkeys`. `notify` identifies no window and needs none of this.

- **tmux (Linux, macOS, WSL):** the pane in `$TMUX_PANE`, and only when that pane's process is an ancestor of the hook. No title is needed. This is the most reliable method.
- **X11 (xdotool):** the one visible window owned by the nearest ancestor process of the hook. If that process owns more than one window, `windowTitle` has to narrow the choice to one. If no ancestor owns a window, exactly one window must match `windowTitle`. The keystroke goes to that window id after it is activated and confirmed active.
- **Windows (sendkeys):** chosen the same way, by owner process first and then by title. Declines outright, before any of that matters, when the resolved window is owned by Windows Terminal — see the method table above. Otherwise, after the delay, the command is sent only when that exact window handle still has focus; if it does not,

nothing is typed and the decline is logged.

Windows Terminal runs every tab in one window. A process-id match cannot tell which tab is active, so on Windows Terminal set `windowTitle` to something only this session's tab shows.